Introduction to machine learning -10-

Probabilistic methods

Maximum likelihood

Let D be the observation from a random process

Let $x^{(i)}$ be independent identically distributed random variables

$D = (x^{(1)}, x^{(2)}, \dots, x^{(N)})$ an observation

e.g. coin flip D: { $N_H = 55$ heads, $N_T = 45$ tails }

Likelihood: probability of observed data

θ: parameter of the model, e.g. prob. of heads

$L(\theta) = p(D) = \theta^{N_H}(1-\theta)^{N_T}$ (sequence not just count)

$L(\theta)$ is very small $L(0,5) = \frac{1}{2^{100}} \approx 10^{-30}$

It is customary to take the log

logarithm: strictly monotonous, conserves the position of the maxim

Example 1.:

$$l(\theta) = \log \mathcal{L}(\theta) = N_H \log \theta + N_T \log(1-\theta)$$

θ_{max} if derivative is zero

$$0 = \frac{dl}{d\theta} = \frac{N_H}{\theta} - \frac{N_T}{1-\theta}$$

$$\theta_{ML} = \frac{N_H}{N_H + N_T} \quad \text{(as expected)}$$

Example 2: $x^{(i)}$ is Gaussian distributed

$$p(x^{(i)}) = \frac{1}{\sqrt{2\pi\sigma}} \exp\left[-\frac{(x^{(i)}-\mu)^2}{2\sigma^2}\right]$$

μ: mean, σ: variance

log likelihood:

$$l(\mu, \theta) = \log \prod_{i=1}^{N} p(x^{(i)}) = \sum_{i=1}^{N} \log p(x^{(i)}) =$$

$$= \sum_{i=1}^{N} \left\{ -\frac{1}{2}\log(2\pi) - \log(\sigma) - \frac{(x^{(i)}-\mu)^2}{2\sigma^2} \right\}$$

maximum:

$$0 = \frac{\partial l}{\partial \mu} = \frac{1}{2\sigma^2} \frac{\partial}{\partial \mu} \sum_{i=1}^{N} (x^{(i)}-\mu)^2 = \frac{-1}{\sigma^2} \sum_{i=1}^{N} (x^{(i)}-\mu)$$

$$\Rightarrow \frac{1}{N} \sum_{i=1}^{N} x^{(i)} = \mu_{ML}$$

$$0 = \frac{\partial l}{\partial \sigma} = \sum_{i=1}^{N} -\frac{1}{\sigma} + \frac{1}{\sigma^3}(x^{(i)}-\mu)^2$$

$$\sigma_{ML}^2 = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (x^{(i)}-\mu)^2}$$

As expected, the mean should be the mean, the var. the var but we have proven that it gives the highest prob.

Likelihood gives the probability of the observations given the parameters ($p(D|\theta)$)

- Log likelihood is a sum for independent events → easy handling → trivial results
- Problems with sparse data:
  eg.: two flips, both heads → $\theta_{ML} = 1$?

Let us reverse $p(D|\theta)$ to $p(\theta|D)$: the probability of the parameters θ given the observation D

Bayesian approach, parameter estimation

Input:

- prior distribution $p(\theta)$, generally not known, replaced by our prior beliefs about θ
- likelihood $p(D|\theta)$ as before

Output:

- posterior distribution

$$p(\theta|D) = \frac{p(\theta)\,p(D|\theta)}{\int p(\theta')\,p(D|\theta')\,d\theta'}$$

↳ parameter probability after observation

- posterior predictive distribution

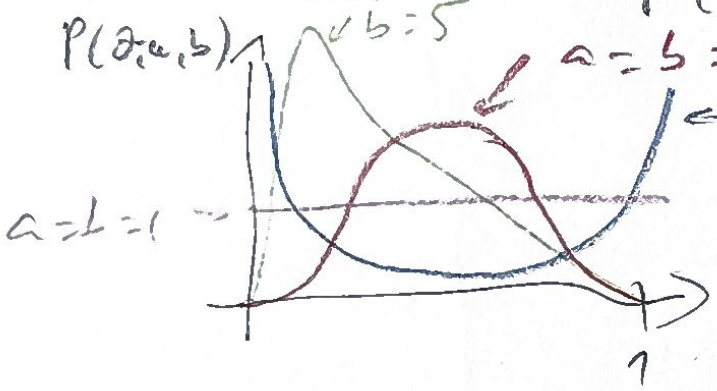
$$p(D'|D) = \int p(\theta|D)\, p(D'|\theta)\, d\theta$$

↖ uncertainty about params

How to choose $p(\theta)$?

- We know that $\theta \simeq 0{,}5 \Rightarrow p(\theta) = \delta(0{,}5-\theta)$
- We do not know anything: $p(\theta) = 1$
- Usual choice: Beda distribution

$$p(\theta | a,b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\, \theta^{a-1} (1-\theta)^{b-1}$$



↓ generalized binomial distr.

$$p(\theta; a,b) \sim \theta^{a-1} (1-\theta)^{b-1}$$

$$p(\theta|D) \propto p(\theta)\, p(D|\theta) = \left[\theta^{a-1}(1-\theta)^{b-1}\right]\left[\theta^{N_H}(1-\theta)^{N_T}\right] =$$

$$= \theta^{a-1+N_H} (1-\theta)^{b-1+N_T}$$

Let us calculate the posterior predictive distribution, i.e. the probability that the next flip is a head

$$p(x'=H|D) : \theta_B = \int p(\theta|D)\, \underbrace{p(x'=H|\theta)}_{\theta}\, d\theta :$$

$$= \int \text{Beta}(\theta, N_H + a, N_T + b) \cdot \theta \cdot d\theta :$$

$$= \frac{N_H + a}{N_H + N_T + a + b}$$

θ_B goes to $\frac{N_H}{N_H + N_T}$ for large number of flips. Prior probability smoothens out the part with few flips

Maximum a-posterior (MAP) approximation

Problem with the full Bayesian is that the integral is not always doable:

$$\theta_{MAP} = \arg\max_{\theta} p(\theta|D) =$$

$$= \arg\max_{\theta} \left[\log p(\theta) + \underbrace{\log p(D|\theta)}_{\text{this was log likelihood}} \right]$$

coin flip:

$$\log p(\theta, D) = \log p(\theta) + \log p(D|\theta) =$$

$$= \text{const} + (a-1)\log\theta + (b-1)\log(1-\theta) +$$

$$+ N_H \log\theta + N_T \log(1-\theta)$$

$$\hat{\theta}_{MAP} = \frac{N_H + a - 1}{N_H + N_T + a + b - 2} \quad \left(\text{Mode of Beta } \frac{a-1}{a+b-2} \right)$$

Example : coin flips

$a = b = 2$

	$N_H = 2, N_T = 0$	$N_H = 55, N_T = 45$
$\hat{\theta}_{ML} = \frac{N_H}{N_H + N_T}$	1	$\frac{55}{100} = 0,55$
$\theta_B = \frac{N_H + a}{N_H + N_T + a + b}$	$\frac{4}{6} \simeq 0,67$	$\frac{57}{104} \simeq 0,548$
$\hat{\theta}_{MAP} = \frac{N_H + a - 1}{N_H + N_T + a + b - 2}$	$\frac{3}{4} = 0,75$	$\frac{56}{102} \simeq 0,549$